



Center for Geographical Information System University of the Punjab



A COMPARATIVE STUDY OF GROUNDWATER QUALITY USING GIS BASED TECHNIQUES FOR ADMINISTRATIVE TOWNS OF LAHORE CITY

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Acknowledgements

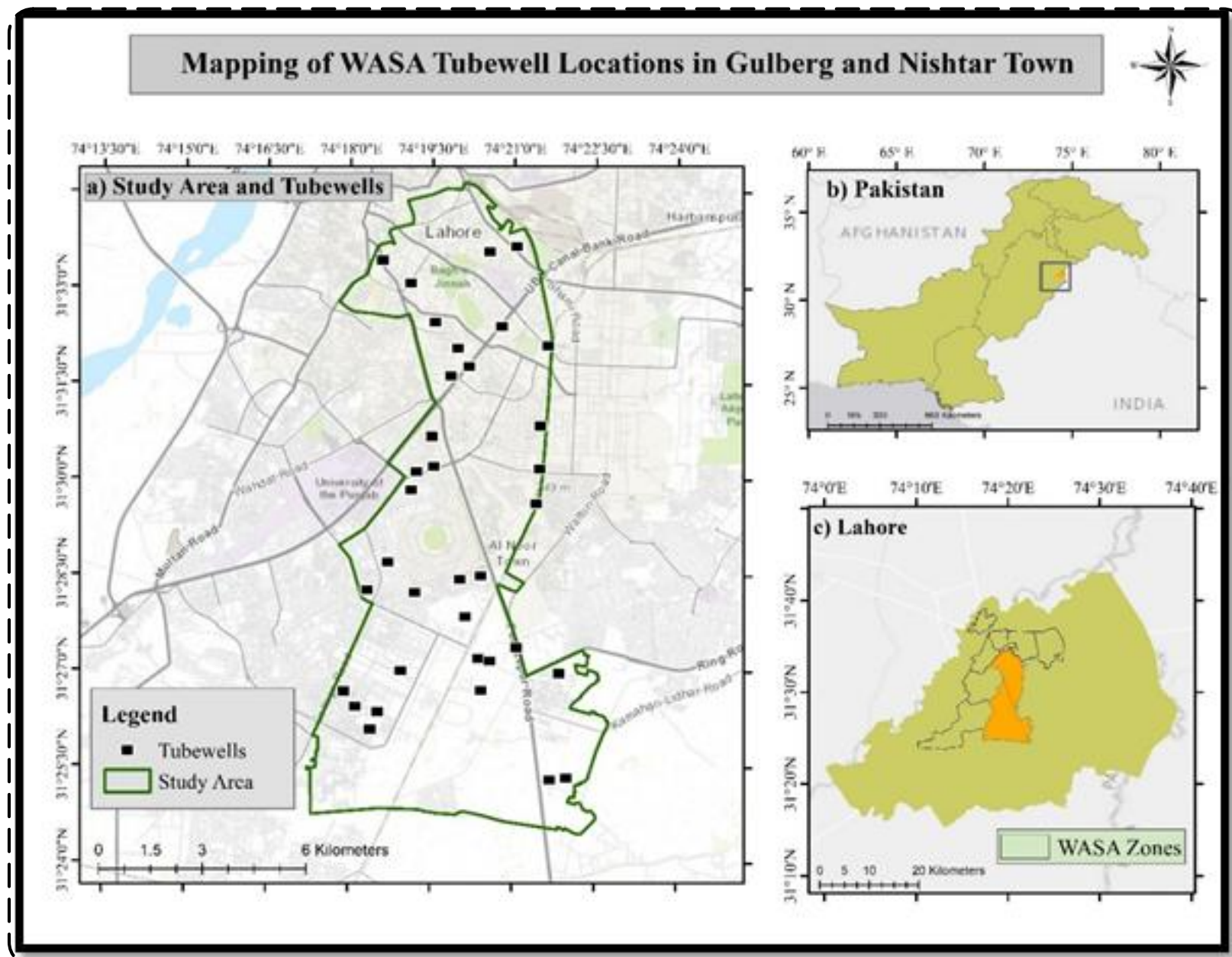


Figure 1. Spatial Distribution of WASA Tubewells in Gulberg and Nishtar Towns. Inset b) and c) place the study area within Pakistan and Lahore, respectively, showing its administrative context and position within WASA zones a) highlights tubewells locations as black squares within the green outlined study area.

Introduction

- Lahore faces groundwater contamination from urbanization, pollution, and sewage, especially in Gulberg and Nishtar.
- GIS and WQI methods (Traditional, Fuzzy, Entropy) were used to assess groundwater trends for 2014, 2019, and 2024.
- Results support WASA, EPA, and health departments in managing water quality and preventing disease.



Research purposes

To evaluate groundwater quality in Lahore using GIS and compare Traditional, Fuzzy, and Entropy-based WQI methods.

Research Problem



Knowledge Gap

Limited studies compare traditional and advanced (Fuzzy, Entropy) WQI methods using GIS to assess spatial-temporal groundwater quality in Lahore (Ramakrishnaiah et al., 2009).



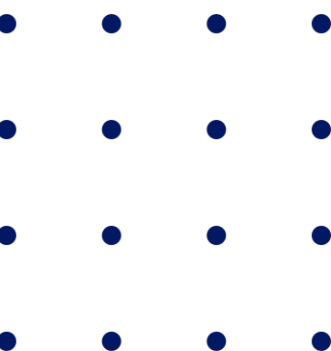
Practical Challenge

Integrating and comparing Traditional, Fuzzy, and Entropy-based WQI methods within a GIS framework is challenging due to their differing assumptions, data sensitivity, and interpretation of uncertainty.



Theoretical Debate

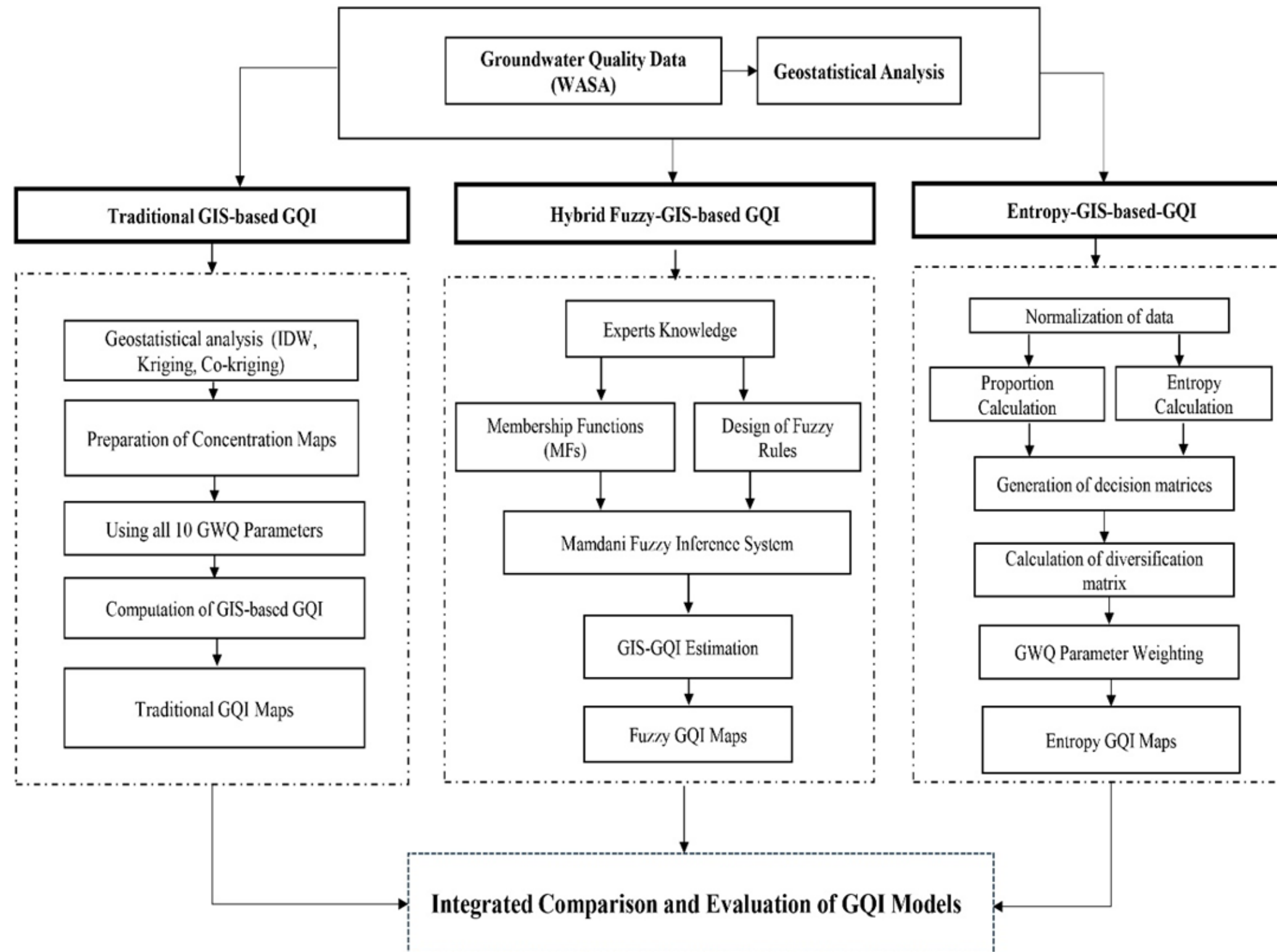
There is ongoing debate among researchers on which method is most effective for groundwater quality assessment, due to differences in their approaches, data interpretation, and handling of uncertainty.



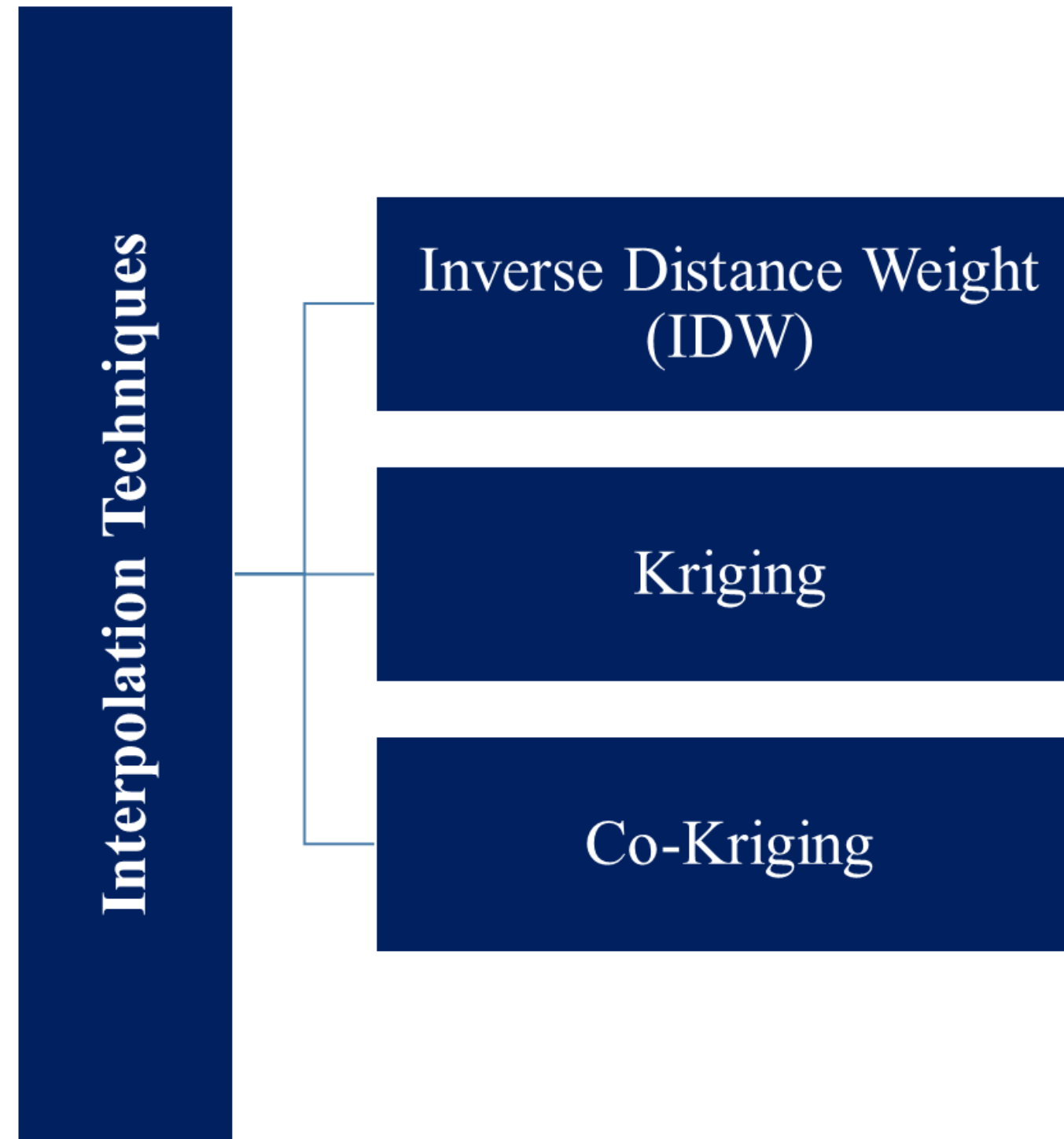
Literature Review

SR No.	Summary	Methods	Authors	Source
1	This study proposes and tests a new Fuzzy-GIS-based index to more accurately assess and map drinking water quality in Southern India.	Fuzzy Logic System and Traditional Based Groundwater Quality (GQI) Indices	• Madan Kumar Jha • Ankit Shekhar • M. Annie Jenifer	https://www.sciencedirect.com/science/article/pii/S0043135420304048
2	This study applies Shannon's entropy and GIS to map and evaluate groundwater quality in the Lower Ganga Basin for better contamination management and sustainability.	Shannon's Entropy Weighted Model	• Mohd Sayeed Ul Hasan • Abhishek Kumar Rai	https://www.sciencedirect.com/science/article/pii/S095965262033122X
3	The thesis evaluates Lahore's groundwater vulnerability using a Modified DRASTIC model in GIS, emphasizing arsenic contamination and guiding water quality management.	Drastic Model	• Syed Umair Shahid	https://www.academia.edu/download/120285835/Syed_Umair_Shahid_Remote_Sensing_Geogr_inf_or_HSR_2017_NUST_4.12.2017.pdf

FLOW CHART



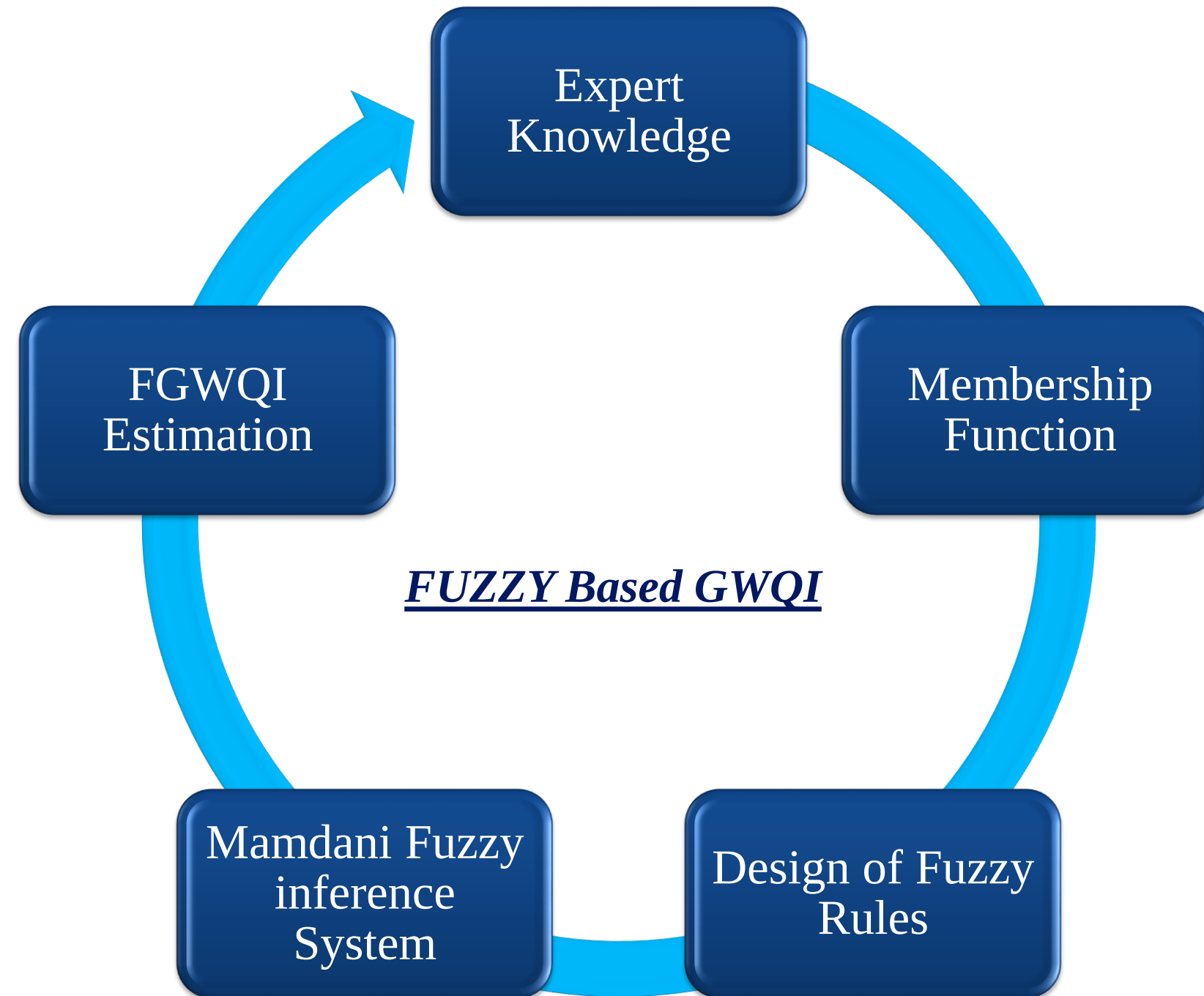
Methodology



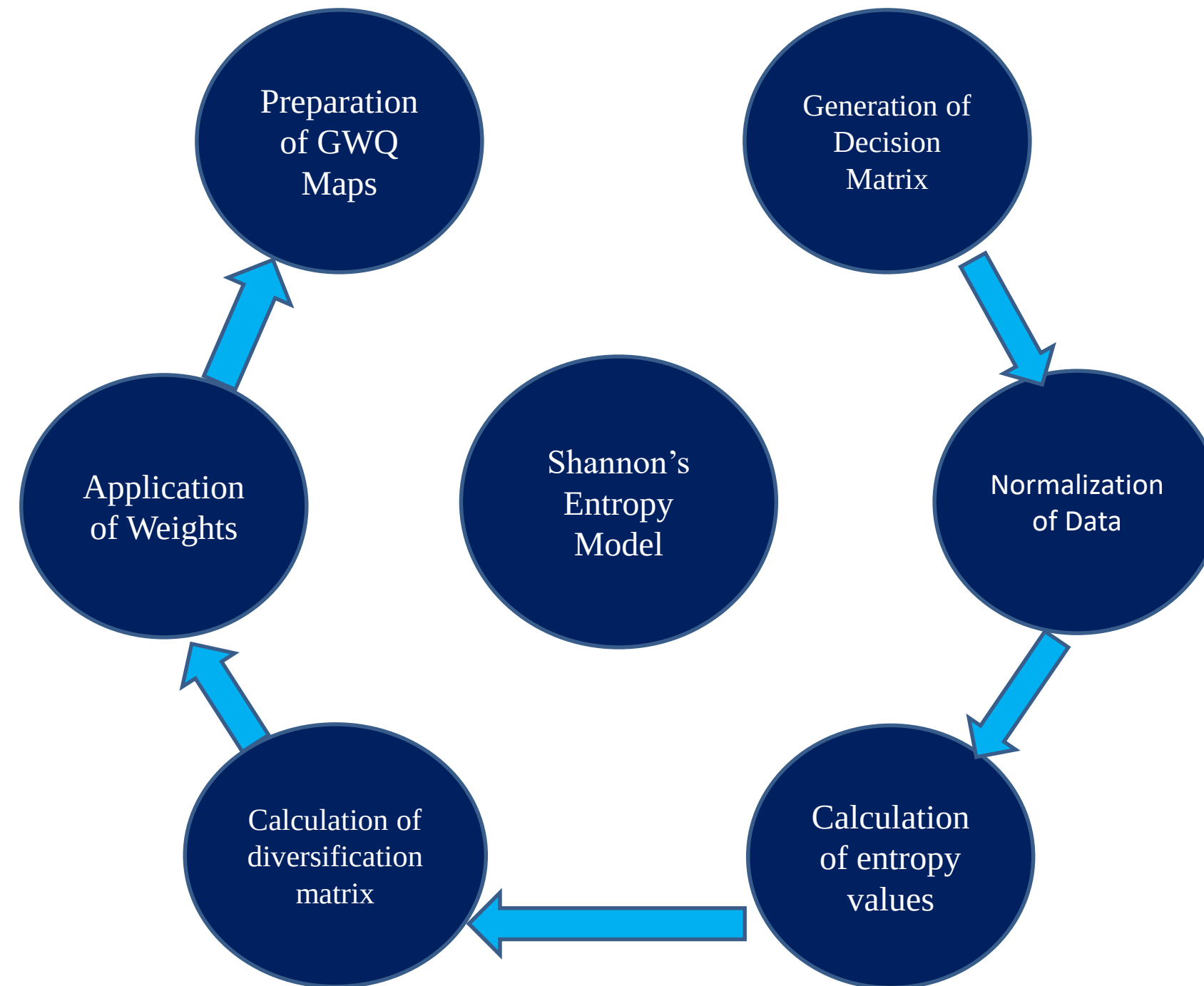
Methodology



Methodology



Methodology



Results & Discussions

TRADITIONAL GWQI

2014:

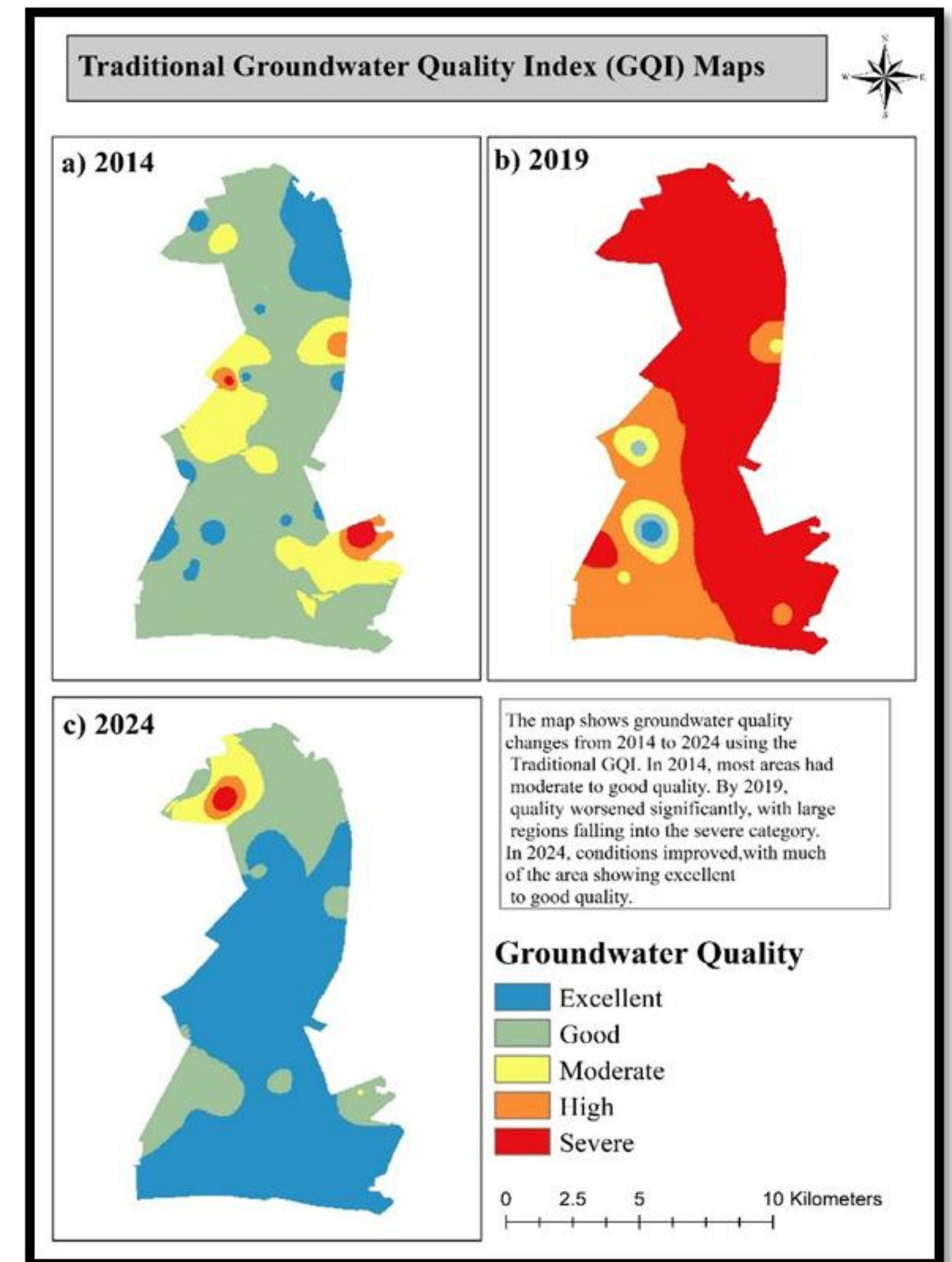
- Groundwater mostly Good to Moderate quality
- Scattered zones of Excellent and few High/Severe areas

2019:

- Sharp decline in quality
- Majority of region turned Severe or High
- Indicates significant pollution or lack of control measures

2024:

- Strong recovery observed
- Dominance of Excellent and Good quality
- Severe zones reduced drastically, limited to north



Results & Discussions

FUZZY Based GWQI

2014:

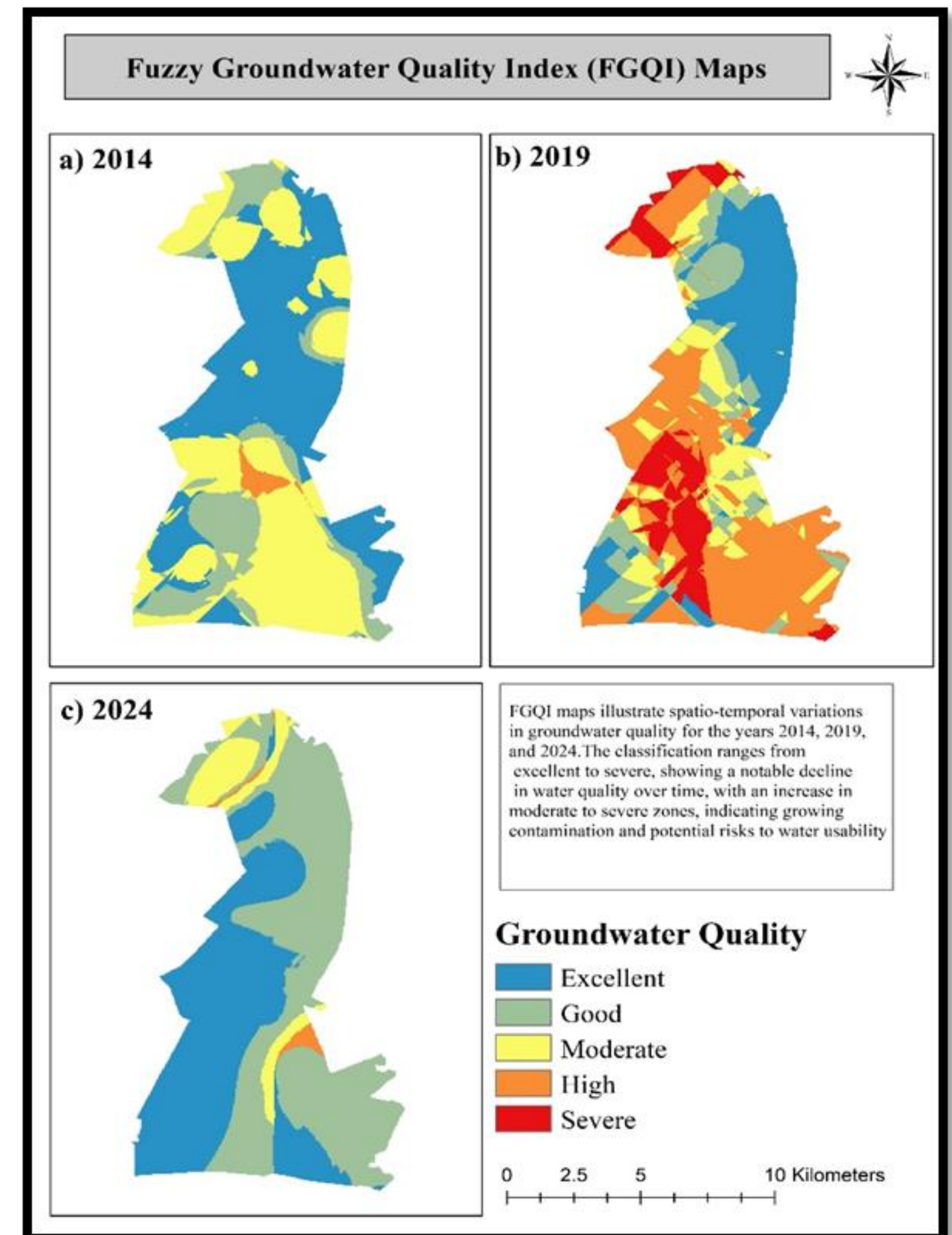
- Majority of areas showed Good to Excellent quality
- Scattered Moderate and High zones observed
- Groundwater generally safe and usable

2019:

- Sharp deterioration in quality, especially in GQI maps
- FGQI also shows increased Severe and High zones
- Indicates rising contamination and poor water management

2024:

- Significant improvement in both indices
- Dominance of Excellent and Good zones
- Limited High/Severe areas, reflecting recovery



Results & Discussions

Shannon's Entropy Model

2014:

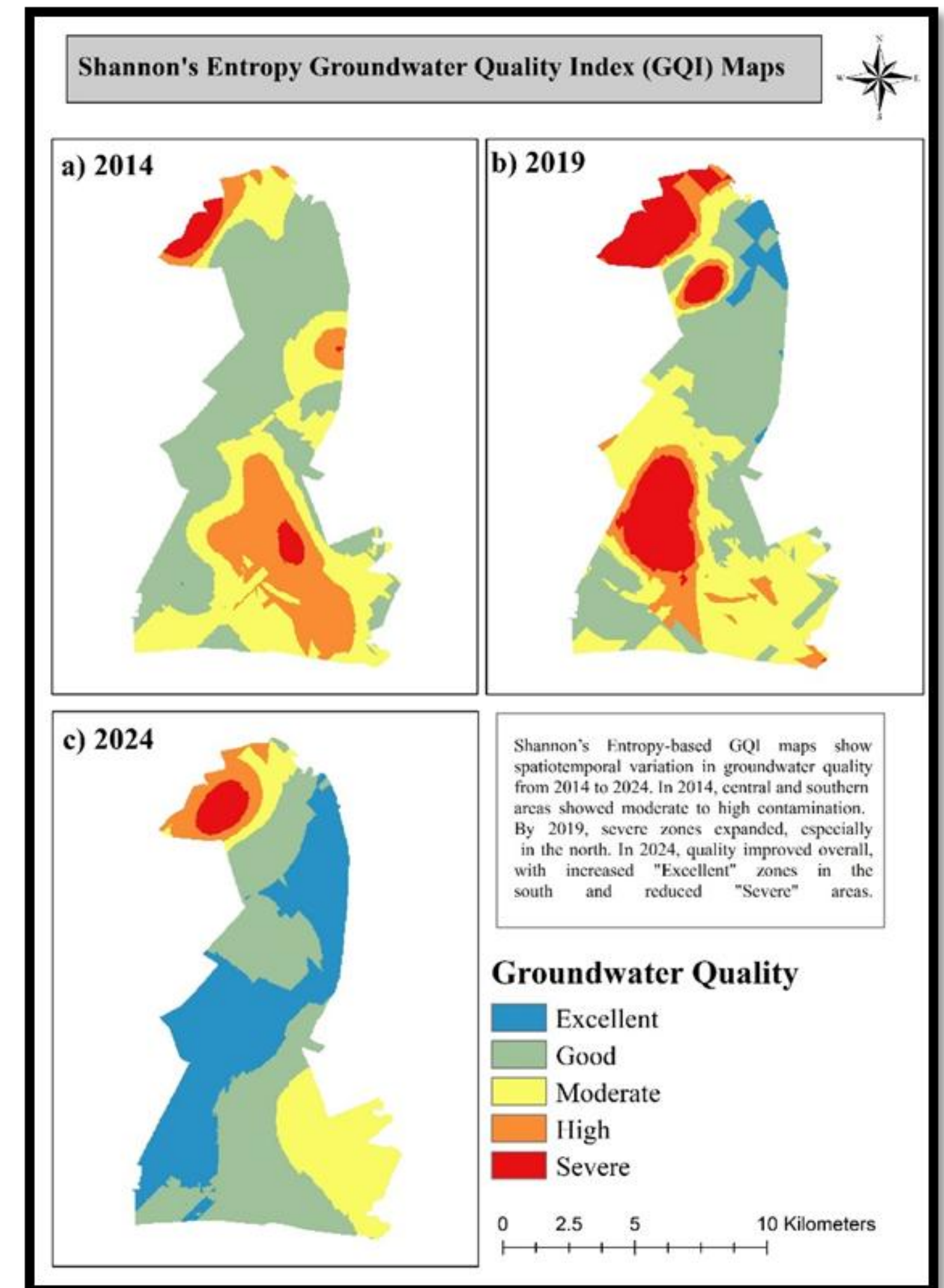
- Central and southern regions had Moderate to High contamination
- Few Severe (red) zones in the north and south
- Most other areas showed Good quality

2019:

- Significant deterioration, especially in northern and central areas
- Large Severe zones expanded, particularly in the center
- Some isolated Excellent areas appeared in the northeast

2024:

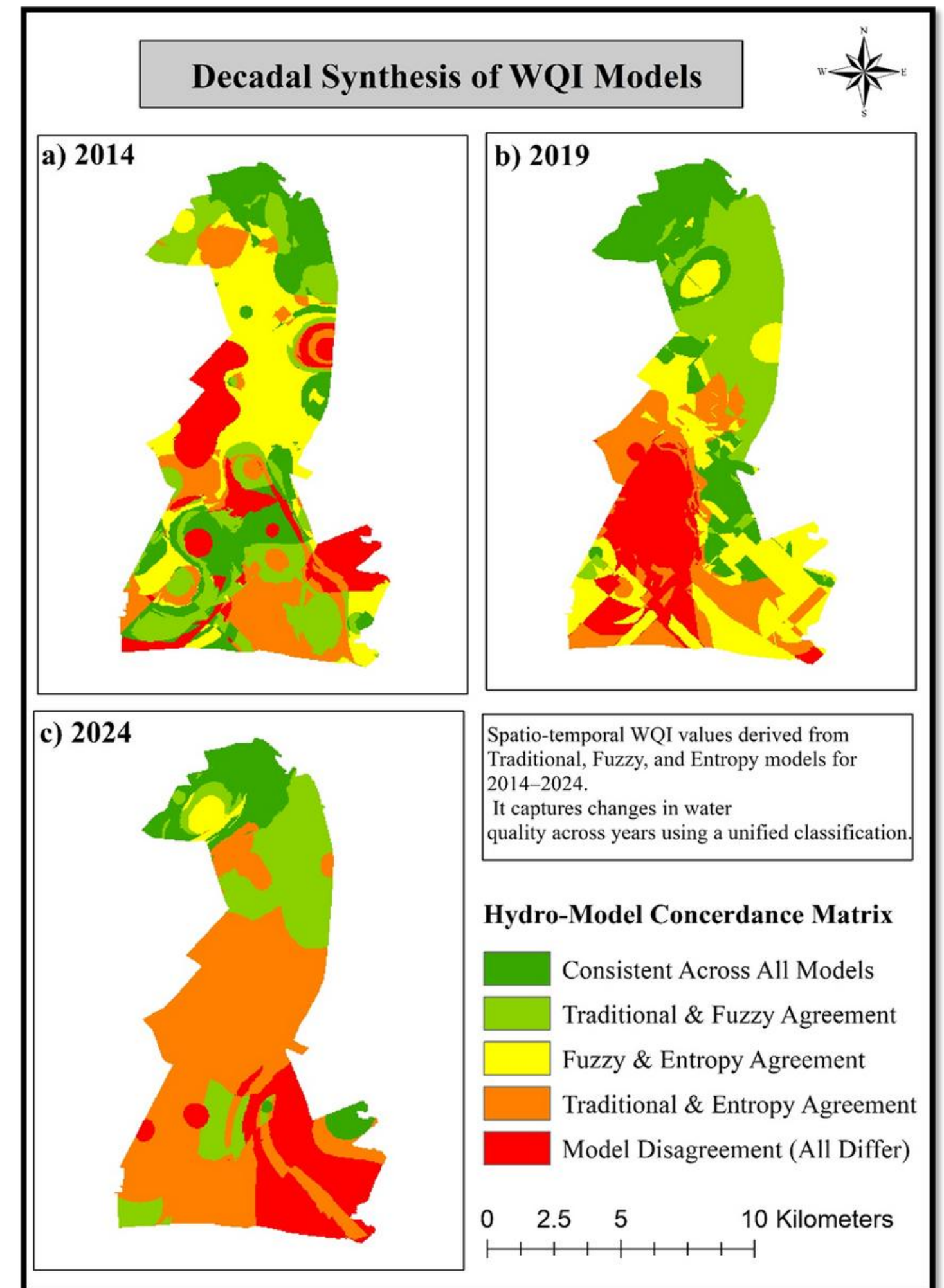
- Noticeable improvement in overall water quality
- Broad spread of Excellent (blue) and Good (green) zones
- Only one major Severe spot remains in the north



Results & Discussions

Comparative Analysis of all Three Models

- Three WQI Models Analyzed: Traditional, Fuzzy Logic, Entropy
- Hydro-Model Concordance Matrix used to assess spatial agreement.
- Entropy model shows balanced and stable groundwater quality patterns over time..
- Traditional model oversimplifies; Fuzzy Logic is sensitive and sometimes inconsistent
- 2014 had major model disagreements; 2019 & 2024 showed improved consistency.
- Entropy WQI recommended for long-term water quality monitoring and policy planning.



Results & Discussions

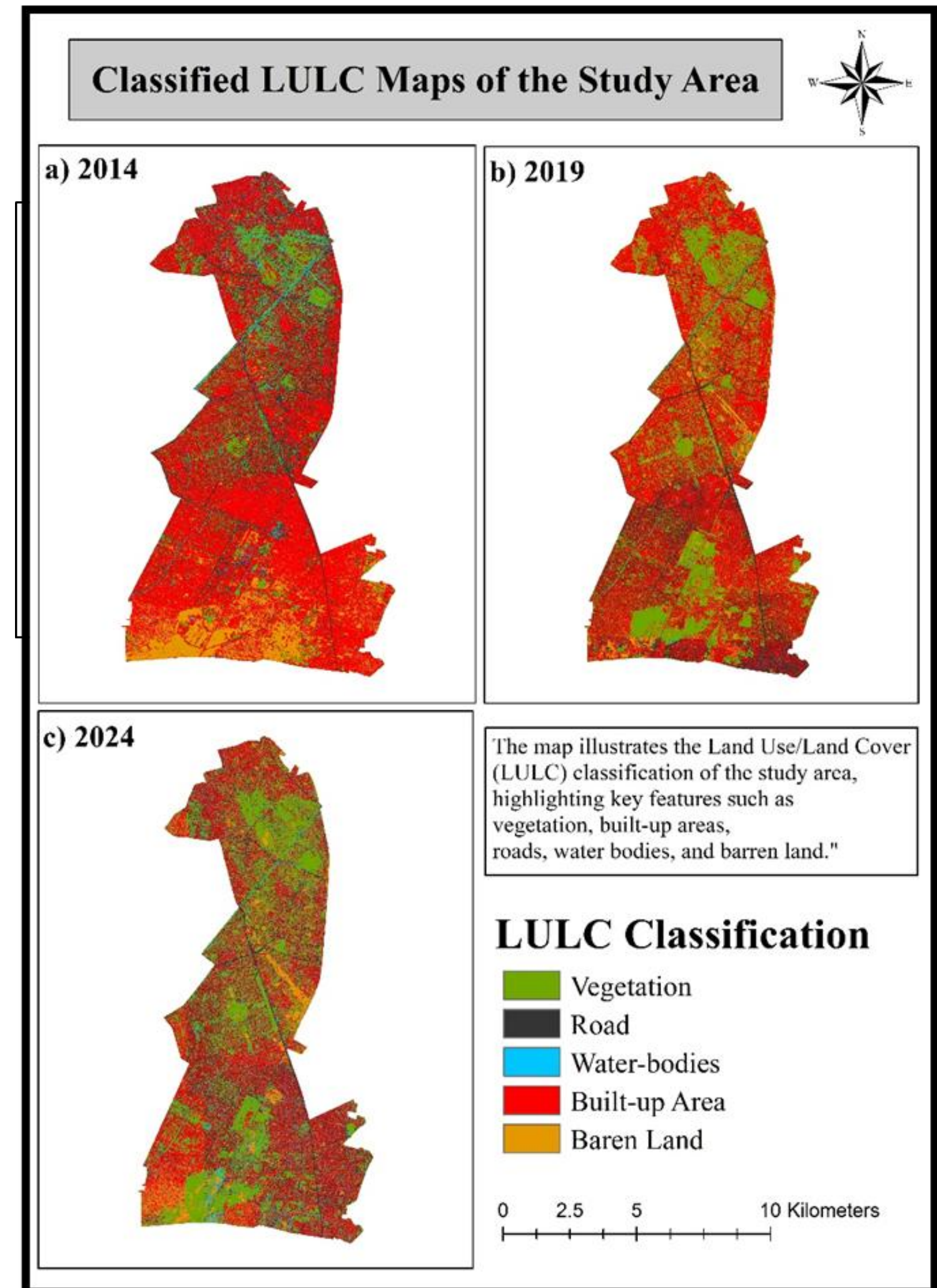
Integration with Rainfall and LULC

Rainfall:

- 2014: Low rainfall, poor recharge, high contamination.
- 2019: More rain, but quality still poor due to leaking sewage.
- 2024: Quality improved with rainfall & WASA infrastructure upgrades.

Landuse Land Cover:

- Vegetation rose significantly (from 3364 to 12108 m²).
- Natural and human factors shaped groundwater and land use.



Conclusion



- □ Significant spatial and temporal variations in groundwater quality were observed across Gulberg and Nishtar towns.
- □ Groundwater deterioration is primarily attributed to unregulated urban growth, industrial discharge, and sewage infiltration.
- □ GIS-based approaches using Fuzzy Logic and Entropy models provided more accurate and flexible assessments than traditional methods.
- □ The study emphasizes the need for targeted policy interventions and enhanced monitoring systems.
- □ Findings support the implementation of sustainable groundwater management strategies to ensure access to safe drinking water.

Acknowledgements



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ANY QUESTION?

THANK YOU